

Understanding Cloud Storage Mechanisms

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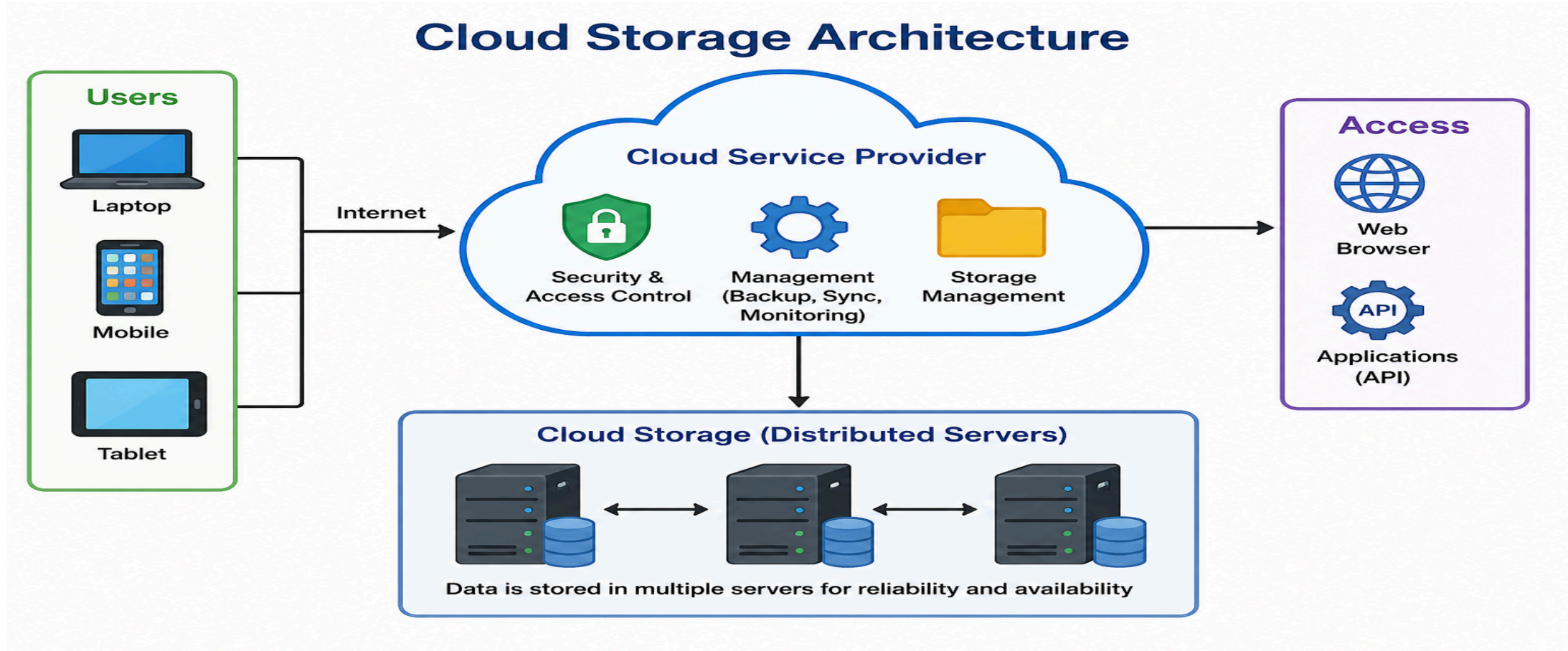
Objectives

- Define **cloud storage** and understand its purpose.
- Explain **how cloud storage stores, manages, and retrieves data**.
- Describe the **basic architecture** of cloud storage.
- Identify the **benefits and industry applications** of cloud storage.
- Understand a **real-world example** of cloud storage, such as Google Drive.

Introduction

- Cloud storage stores data on **remote servers** instead of local devices.
- Users can **access data anytime and anywhere** using the internet.
- It provides **secure storage** with encryption and backup.
- Storage capacity can be **easily increased** as needed.
- It is widely used by individuals and businesses for **file storage, sharing, and backup**.

Cloud Storage Architecture



Working of Cloud Storage

- Users upload files through the internet.
- Data is sent to the cloud service provider.
- Files are stored across multiple servers.
- Data is encrypted and backed up for security.
- Users can access files anytime using authenticated devices.
- Synchronization keeps files updated across all connected devices.

Types of Cloud Storage

1. Object Storage

- Stores data as objects with metadata.
- Best for images, videos, backups, and large files.
- Highly scalable.

2. File Storage

- Organizes data in folders and files.
- Easy to share and access.
- Commonly used for documents and team collaboration.

3. Block Storage

- Divides data into fixed-size blocks.
- Provides high performance and low latency.
- Used for databases and virtual machines.

Practical Example

Google Drive

- User uploads a file to **Google Drive**.
- The file is stored securely on Google's cloud servers.
- Data is backed up and replicated across multiple servers.
- User can access the file anytime from a phone, laptop, or tablet.
- Changes made on one device are automatically synchronized to all other devices.

Advantages & Challenges

Advantages

 Access data from anywhere

 Easy to scale storage

 Cost-effective (pay-as-you-use)

 Automatic backup and recovery

 Easy file sharing and collaboration

Challenges

 Requires an internet connection

 Privacy and security concerns

 Possible network latency

 Service outages may affect access

 Dependence on the cloud provider

Conclusion

- Cloud storage provides **secure, scalable, and reliable** data storage over the internet.
- It enables users to **store, access, and share files anytime, anywhere.**
- Features such as **backup, synchronization, and encryption** improve data protection and availability.
- Despite challenges like **internet dependency and security concerns**, cloud storage remains an essential solution for individuals and businesses.

THANK YOU